

The operating system for maintenance intelligence.

Turning shift handovers into structured data, machine memory, and predictive intelligence — for mining, heavy commercial, and infrastructure sites.

Prepared for: **Casey** · Founder · ShiftForge

Delivered by: **Arc** · via **Earthed Group** (Caleb Vears)

Related entity: **Top Gun Engineering** · delivery + integration partner

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01 · THE PROBLEM WORTH SOLVING

Every mine site + heavy commercial maintenance operation in Australia runs on the same broken model:

- **Fitters fill in paper (or basic digital) shift handover sheets** that supervisors never read carefully
- **Machines fail** because the last three shifts flagged the same problem and nobody joined the dots
- **Enterprise systems** (SAP-PM, Pronto, HxGN EAM, Cammar) cost \$2M+ to deploy and don't talk to the crews on the tools
- **Institutional memory** lives in the heads of long-serving fitters — when they retire or move sites, decades of knowledge walk out the door
- **Cross-shift coordination** falls apart at 6am handover — day crew inherits chaos, wastes 60-90 min working out what's still open
- **Repeat faults** aren't flagged as patterns until the equipment fails — costing tens of thousands per unplanned downtime hour

The cost of this is staggering. Mining downtime is \$5,000-\$50,000/hour depending on the asset. A typical mid-tier mine burns 8-15% of its total maintenance budget on preventable rework caused by lost information between shifts. That's \$2-6M/year on a \$50M ops budget.

ShiftForge doesn't replace SAP-PM. It sits BETWEEN the crews on the tools and the enterprise systems above — making the shift handover the smartest touch-point in the entire maintenance workflow.

02 · THE FOUR-LAYER ARCHITECTURE

ShiftForge is built as four stacked layers, each solving a specific failure mode of current-state mining maintenance.

Layer 1 · Asset Register

The memory. Every machine gets a unique digital twin with permanent history.

Layer 2 · Shift Handover Engine

The handshake. Every shift starts + ends with structured briefing + report, tied to specific assets.

Layer 3 · Intelligence Layer

The brain. AI-driven report generation, pattern detection, predictive flags, natural-language querying.

Layer 4 · Personnel + Permit Integration

The guardrails. Role-based access, permit-to-work sync, competency verification, escalation ladder.

03 · LAYER 1 · ASSET REGISTER (THE MEMORY)

Every machine + every piece of equipment on site gets a unique ID and a permanent, structured digital record — not spreadsheet rows. This is the foundation. Without it, ShiftForge is just another form-filler.

Each asset record holds:

- ✓ Make, model, serial number, install date, expected service life
- ✓ Location hierarchy — Site → Level → Area → Position
- ✓ Discipline classification — Mechanical / Electrical / Hydraulic / Pneumatic / Structural / Instrumentation
- ✓ Ownership + custodian crew (which team is responsible)
- ✓ Manufacturer specs, service manuals, spare parts BOM (Bill of Materials)
- ✓ Warranty status + contract terms
- ✓ Full lifetime history — every shift log, every fault, every part replaced, every service performed
- ✓ Health score (calculated from fault frequency + wear indicators + service adherence)
- ✓ Attachments — photos, drawings, service manuals, warranty docs

The killer feature

A fitter selects "C32 Conveyor" from a dropdown and ShiftForge instantly knows: it's a Cat C32 conveyor drive, it's been running for 4,721 hours, last service was 3 shifts ago, the previous crew flagged a bearing temp anomaly on shift 4,715, the recommended action was follow-up thermal scan, and the on-site spare bearing is in Bay 3. All before the fitter has typed a single character.

No mixing up. Each equipment ID has its own memory silo. C32-01 is different from C32-02 even if they're the same model. Every data point lands against the right machine. Every dropdown reads from the Asset Register — no free text where structured data exists.

04 · LAYER 2 · SHIFT HANDOVER ENGINE (THE HANDSHAKE)

The core workflow that ShiftForge orchestrates for every shift, every day, on every site.

Shift start · auto-generated Handover Briefing

- ✓ Read + acknowledge required before the crew can lodge new work
- ✓ Summarises the outgoing shift's report + open items + AI-detected trends
- ✓ Escalates anything unresolved for 2+ shifts to the supervisor
- ✓ Highlights safety-critical alerts + permits still active
- ✓ Role-filtered view: mechanic sees mechanical items, sparky sees electrical, etc.

Shift end · Report generation flow

- ✓ Fitter opens the shift report form (Casey's existing MVP · upgraded UX)
- ✓ Dropdowns tied to Asset Register — no free-text where structured data can be captured
- ✓ Location cascades: Site → Level → Area → Position
- ✓ Equipment is searchable multi-select from Asset Register
- ✓ Work Completed / Issues / Actions / Parts / Delays / Status / Planned Next / Recommendations — text fields but each auto-tagged to the selected equipment(s)
- ✓ AI Draft button — turns form fields into a polished professional report in the client's writing voice
- ✓ Fitter reviews AI draft, edits, signs off with role-based auth
- ✓ Submit → auto-distributes to next shift + supervisor + logs to each asset's history + updates health scores

05 · LAYER 3 · INTELLIGENCE LAYER (THE BRAIN)

This is where Casey's "AI hold point" gets solved. Layered intelligence, not one big AI feature:

Immediate (real-time)

- **Report generation** — Claude / GPT via API turns form fields into a polished report in the client's voice
- **Auto-tag equipment references** in free-text fields (fitter mentions "the C32 vibration" → ShiftForge auto-links to C32-01)
- **Instant retrieval** — supervisor asks "what's the story on C32-01" and gets full history

Weekly

- **Pattern detection** — flags equipment with repeat faults, rising fault frequency, correlating events across assets
- **Cross-shift theme summaries** — "3 hydraulic faults on level 72 in last 5 shifts — investigate common root cause"
- **Parts consumption trends** — "conveyor bearing consumption up 40% this quarter · investigate supplier quality"

Monthly

- **Predictive maintenance flags** — "C32-01 is trending 30% faster wear than fleet average · investigate before scheduled service"
- **Fleet benchmarking** — anonymised cross-site data (opt-in) shows how a site's equipment compares to industry averages
- **Cost-avoidance reporting** — "\$X in downtime prevented by early flags this month"

On-demand

- **Natural language query** — "show me every hydraulic fault on level 72 in the last 60 days" → summary + report

■ **Cross-shift memory** — "this issue was raised 3 shifts ago by [name], the fix was [action], the recommended follow-up was [work] · here's the latest status"

06 · LAYER 4 · PERSONNEL + PERMIT INTEGRATION (THE GUARDRAILS)

This is where ShiftForge crosses into eSwitching + ePermits territory — turning a maintenance tool into a safety + compliance system.

- ✓ **Role-based views** — mechanic sees mechanical assets + relevant history; sparky sees electrical + isolations; supervisor sees everything + escalations; engineer sees trends + recommendations
- ✓ **Permit-to-work integration** — every action logged is tied to the active permit; ShiftForge knows if work happened under valid isolation
- ✓ **Isolation register sync** — pulls from the client's isolation system (Ampac, Innovaccer, or homegrown) so ShiftForge knows what's live vs dead
- ✓ **Competency check** — before a fitter can sign off "hydraulic pressure test done," ShiftForge verifies their ticket is current + valid for that scope
- ✓ **Incident escalation ladder** — critical fault types auto-escalate through defined chain (fitter → supervisor → engineer → mine manager)
- ✓ **Audit trail** — every data point captured with user + timestamp + geo-tag (if enabled) — meets mining regulator record-keeping requirements
- ✓ **Time on tools tracking** — ShiftForge captures actual work time vs delays (permits waiting, isolations waiting, parts waiting) — feeds directly into productivity KPIs

07 · THE TOP GUN ENGINEERING DELIVERY MODEL

ShiftForge is the SaaS product. Top Gun Engineering is the humans who deploy it into real mine sites and heavy commercial operations. This dual structure is deliberate — SaaS revenue is high-margin recurring, consulting revenue is high-fee upfront. Both feed each other.

Top Gun Engineering delivers:

- ✓ **Site scoping + Asset Register buildout** — 2-4 week engagement to load the client's fleet into ShiftForge, verify serials, capture location hierarchy, upload service manuals
- ✓ **Crew training** — hands-on sessions with fitters, supervisors, engineers · role-specific workflows
- ✓ **Integration engineering** — connect ShiftForge to the client's existing systems (SAP-PM, Pronto, HxGN EAM, isolation registers, ERP)
- ✓ **Ongoing consulting** — monthly retainer for pattern analysis reviews, roadmap sessions, expansion planning
- ✓ **Custom development** — client-specific integrations, custom reports, bespoke dashboards

Why this structure works:

Enterprise mining clients don't buy SaaS from a startup. They buy delivered outcomes from a consulting firm. Top Gun Engineering is the trusted advisor + implementation partner. ShiftForge is the tech they hand over during deployment. The client sees a proven consulting firm, backed by proprietary tech.

08 · BUSINESS MODEL + PRICING

Three revenue streams, layered:

Stream	Structure	Indicative pricing	Margin
ShiftForge SaaS	Per-user per-month · site licences for 50+ users	\$8k-15k/user/mo · \$8k-25k/mo site plans	70-85%
Top Gun Implementation	Site deployment + integration + training	\$30k-80k per site · one-off	50-65%
Top Gun Consulting	Ongoing retainer for optimisation + expansion	\$8k-15k/mo per client · rolling 12mo	60-75%
Data services (future)	Anonymised fleet benchmarking as an industry 200k+ projects subscriber	\$50k-200k/project	80-90%

First-year revenue potential (5 sites)

5 mine sites × \$50k avg implementation = **\$250k Top Gun** · 5 sites × \$15k/mo SaaS = **\$900k ARR by end of year 1** · 3 clients on \$10k/mo consulting retainer = **\$360k/yr**. Total year 1 potential: **\$1.5M+** at healthy margin.

09 · VERTICAL EXPANSION PATH

Mining first — it's the pain point Casey knows, and the wallet is deepest. Then the same platform scales into adjacent verticals with minor customisation:

- **Mining (Year 1)** — heavy diesel, mechanical fitting, hydraulic systems, conveyors, drills, crushers, loaders, haul trucks
- **Heavy commercial building maintenance (Year 2)** — data centres, hospitals, universities, high-rise commercial (chillers, HVAC, generators, elevators, BMS)
- **Utility asset management (Year 2-3)** — substations, transformers, switching gear, protection relays — direct crossover with Earthed HV territory
- **Renewable energy O&M; (Year 3)** — wind farm turbines (crossover with Vestas / Elecnor world), solar farm plant, BESS installations, hybrid facilities
- **Marine + defence (Year 3-4)** — ship maintenance, port operations, defence base infrastructure

10 · 90-DAY ROADMAP · FROM MVP TO FIRST PAYING SITE

Days 1-30 · Foundation

- ✓ Formalise Asset Register data model (spreadsheet OK for MVP)
- ✓ Wire the "Generate Report" button to Claude API (single call, structured prompt using form fields)
- ✓ Add Equipment dropdown fed from Asset Register (not free-text)
- ✓ Add Location cascade (Site → Level → Area)
- ✓ Persist reports to database (Supabase or Postgres) with per-asset foreign key
- ✓ Deploy to production URL (Vercel + Supabase = live in a weekend)

Days 30-60 · Intelligence

- ✓ Historical report retrieval per asset
- ✓ Handover briefing generator (reads previous shift's report + open items)
- ✓ Simple pattern flags (repeat faults on same asset)
- ✓ Role-based user auth (mechanic vs supervisor vs engineer)
- ✓ Health scoring algorithm (fault frequency × severity + service adherence)

Days 60-90 · Integration + First Sale

- ✓ Pilot site secured (via Top Gun Engineering's existing mining relationships)
- ✓ Free 30-day trial for 1 crew on 1 shift
- ✓ Data export to client's existing systems (CSV to start, API later)
- ✓ Case study built from pilot
- ✓ Second site closed on pilot success

11 · TECH STACK RECOMMENDATIONS

Casey's already on Next.js — solid choice. Round it out with:

Layer	Recommendation
Frontend	Next.js 14 (App Router) + Tailwind CSS + shadcn/ui components
Backend / API	Next.js API routes + tRPC for type-safe endpoints
Database	Supabase (PostgreSQL + Auth + Realtime + Storage) — MVP to enterprise-ready
AI	Claude API (Sonnet for report generation · Opus for pattern analysis) · fallback to GPT-4
Auth	Supabase Auth + role-based access via Postgres Row Level Security
Hosting	Vercel (frontend + API) + Supabase Cloud (database)
File storage	Supabase Storage or Cloudflare R2 for equipment manuals, photos
Analytics	PostHog (product analytics) + Sentry (error monitoring)
Payments	Stripe (subscription billing per site)
Mobile	PWA-first · React Native (Expo) later for offline-heavy mine environments
Voice input	Whisper API for voice-to-text (fitters with dirty hands)

12 · INVESTOR / PARTNER PITCH SUMMARY

The one-liner for anyone Casey talks to about ShiftForge — a mine ops manager, a heavy industry MD, a Top Gun Engineering strategic partner, or a first-cheque angel investor:

"You've got fitters filling out paper handover sheets that supervisors never read. Machines fail because the last three shifts flagged the same problem and nobody joined the dots. Enterprise systems cost \$2M and don't talk to the guys on the tools. ShiftForge is the layer in between — smart enough to remember every machine, every shift, every fitter, and to tell you which asset is about to bite you 240 hours before it does. Deployed by Top Gun Engineering onto your site in 30 days. Paid off inside 3 months on avoided downtime."

Ask

For pilot clients: 30-day free trial · 1 crew · 1 shift · your existing paper process runs in parallel · you decide after 30 days.

For partners: co-deliver on your existing mining contracts · rev-share on SaaS · badge co-branding available.

For investors: \$500k seed round for 12 months runway · target 8 paying sites by month 12 · \$1.5M ARR trajectory · vertical expansion into heavy commercial + utilities from month 18.